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Efficacy of aqueous extract of *Vernonia amygdalina* leaf against strongyle and coccidia infections in sheep

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Abstract

Gastrointestinal parasites are developing resistance to various commercial anthelmintics. Hence, the need to explore the efficacies of herbal plants against gastrointestinal parasites. The study was therefore conducted to determine the efficacy of aqueous extract of *Vernonia amygdalina* leaf against strongyles and coccidia spp. in sheep. A total of 60 sheep were used for the study. The animals were randomly assigned to one of the four groups. Those in group A, B, C, and D were given albendazole (ABZ), 10% aqueous *V. amygdalina* extract (BL10), 20% aqueous *V. amygdalina* extract (BL20), and 10 mL distilled water, respectively. All four treatments were given orally, depending on the body weight of the animals, with the exception of the distilled water, which was constant (10 mL) for each animal. Faecal samples were collected from each sheep and examined using McMaster technique. The data were analyzed using R version 4.2. The study brought to light that the prevalence of strongyles in sheep was 70% in the study area at pre-treatment. BL10 had efficacies of 52.58% and 65.08% against strongyles and coccidia spp., respectively. BL20 produced similar anthelmintic effect against strongyles as ABZ. Strongyle spp. showed resistance against ABZ in the study area.

Key words: helminths, *Vernonia amygdalina*, efficacy, sheep, albendazole

Introduction

Small ruminants contribute to landless, rural farming, peri-urban, and increasingly urban households (Baah et al. 2012) by providing food, income, sociocultural wealth, and clothing (Pollott and Wilson 2009). Sheep provide high quality, balanced, and highly bioavailable protein (Cordeiro et al. 2022) and numerous critical micronutrients, which help achieve a nutritionally balanced diet (Reynolds et al. 2015; Leroy et al. 2022). However, sheep harbour a variety of gastrointestinal (GIT) parasites. GIT parasites have been a major challenge that hinder the production of sheep (Le Bodo et al. 2020). These parasites cause severe weight loss, reproduction failures (Alawa et al. 2001), diarrhea, mortalities (Hansen and Perry 1994; Chartier and Paraud 2012; Garcia-Bustos et al. 2019), and reduction in wool growth (Zajac 2006). Sheep in Sagnarigu district are mostly exposed to these GIT parasites, since most of these animals are reared under extensive system (Salifu et al. 2022), which is characterized by animals scavenging around for feed with no veterinary intervention. These parasites possess threat to the production of the sheep (Maqbool et al. 2017; Halvarsson and Höglund 2021). GIT parasites cause detrimental effects on animal health (Luscher et al. 2005), leading to clinical and subclinical diseases, which results in financial loss and overall decreased productivity (Rahmann et al. 2002).

Ovine coccidiosis is an economically important disease in various parts of the world, which results in reduced productivity due to the associated morbidity and mortality issues (Etsay et al. 2020).

Control of GIT parasites is mainly based on the use of chemical anthelmintic drugs (Castagna et al. 2019; Fissiha and Kinde 2021). However, these are very expensive and not readily available to farmers in rural areas like Sagnarigu. As well as the continuous development of anthelmintic resistance (AR) (Zirintunda et al. 2022) to various chemical anthelmintic compounds and classes as well as their residue and toxicity problems in meat (Kaemmerer and Butenkotter 1973), has necessitated the need for herbal anthelmintics.

Herbal anthelmintics are ethnoveterinary medicinal plants that can expel, stunt growth, or kill helminths (Rehman et al. 2022) when administered to an animal without causing harm to the host. They do so by disrupting the metabolic activities of the helminth, resulting in paralysis and death of the helminths (Nadler et al. 2021). The ethnoveterinary remedies provide a cheaper (Rafique Khan et al. 2021), easily accessible (Xiong and Long 2020), and more sustainable alternative to synthetic drugs and pharmaceuticals (Chakale et al. 2021).

Various herbal anthelmintics have been reported and they include *Aloe barbadensis*, *Capsicum annuum*, *Carica papaya*, *Curuna longa*, and *Vernonia amygdalina*. The *V. amygdalina*

commonly called bitter leaf (English) due to its bitter taste (Farombi and Owoeye 2011; Garba and Oviosa 2019; Gasaliyu et al. 2022), was named after an English botanist William Vernon (Alara et al. 2017). *Vernonia amygdalina* is widely described by livestock farmers as a choice of anthelmintics (Alawa et al. 2008). Luckily, the shrub plant is readily available in most parts of Ghana. The *V. amygdalina* has some bioactive compounds responsible for its ethnobotanical uses (Oyeyemi et al. 2018) such as saponins, alkaloids, flavonoids, terpenes, lignans, phenolic acids, steroids, anthraquinone, coumarins, sesquiterpenes, xanthenes, and edotides (Muraina et al. 2010).

To the best of our knowledge, limited studies have been conducted to assess the efficacy of *V. amygdalina* against GIT parasites in sheep, but none of these studies have been conducted in Ghana. Hence, the study was conducted to determine the efficacy of aqueous extract of *V. amygdalina* leaf against strongyles and coccidia infections in sheep.

Materials and methods

Study area

The study was conducted in the Sagnarigu district, Northern Region of Ghana. Three settlement areas (Nangbagu, Kasalugu, and Shishegu) within the Sagnarigu district were conveniently selected for the study. The district, which has its capital as Sagnarigu, is one of the six districts created in the Northern Region in 2012. It was created out of the Tamale Metropolis by Legislative Instrument (LI) 2066. The Sagnarigu district has 79 communities, comprising of 6 peri-urban, 20 urban, and 53 rural areas (GSS 2010). The district covers a total land size of 200.4 km² and shares boundaries with the Tamale Metropolis to the south and east, Savelugu-Nanton Municipality to the north, Kumbungu district to the north-west and Tolon district to the west. Geographically, the Sagnarigu district lies between latitudes 9°16' and 9°34'N and longitudes 0°36' and 0°57'W (GSS 2010).

Study population, design, and sample size

The study population includes all sheep older than 6 months, which were kept under semi-intensive system, and grazing on natural pasture. This study employed a randomized controlled trials to attain the efficacy of *V. amygdalina* in sheep. A total of 60 sheep, consisted of 16 males and 44 females, that were not given any known anthelmintic for at least 4 months before the commencement of the study, were used. Sheep with no permanent teeth were classified as young sheep, while those who were having at least a pair of permanent teeth were classified as adult sheep.

Preparation of *V. amygdalina* extract mode of application and the dose of the treatment

Fresh leaves of *V. amygdalina* were obtained from the plants at selected areas in Sagnarigu. The leaves were washed to do away with any extraneous materials using tap water. And the leaves were shade-dried for 96 h before they were then

converted into powder using wooden masher and earthenware. Distilled water (Dw), obtained from the UDS Spanish laboratory was then added to the powdered *V. amygdalina* leaf extract. Two concentrations (10% and 20%) of the extract were made as 50 and 100 g/500 mL, respectively. The aqueous extracts were then stored in fridge for 24 h before administering to the required animals.

Both *V. amygdalina* 10% and 20% were given at a dose of 5 mL per 10 kg body weight. Albendazole (ABZ) at 2 mL per 10 kg body weight and Dw at 10 mL per animal. Twenty sheep from each of the three localities were used. Each locality had four treatment groups. The sheep used for the study were randomly assigned to one of the four treatment groups (A, B, C, and D). Those in group A, B, C, and D were given ABZ, 10% aqueous *V. amygdalina* extract (BL10), 20% aqueous *V. amygdalina* extract (BL20), and 10 mL Dw. All four treatments were given orally, depending on the body weight of the animals, with the exception of the Dw, which was constant for all the sheep taking that intervention.

Faecal samples collection

Faecal samples were collected from each of the 60 sheep for the study, at pre-treatment (Wk0) and post-treatment (Wk1 and Wk2). Each animal was given a unique tag before the pre-treatment for easy individual identification. Hand gloves were used and a small amount of water-based lubricant was applied to the index and middle fingers. The index and middle fingers were then inserted into the rectum of each animal and removed approximately 4 g of faecal matter from them. The faecal matter from each sheep was then kept in a faecal sample container and well labelled for easy identification. The samples were then kept in an ice chest and transported to the UDS Spanish Laboratory for analysis using McMaster technique. The samples were analyzed within 12–24 h after arrival at the laboratory.

Faecal eggs/oocyst count

McMaster technique was used to analyze the faecal samples for strongyle eggs and coccidian oocysts as previously described by Hansen and Perry (1994). All the strongyle eggs and coccidian oocysts found within the engraved areas of both chambers were counted and recorded. The total strongyle eggs and coccidian oocysts within both chambers were then multiplied by 50 to obtain the strongyle egg/coccidian oocyst count per gram of faeces (Sandoval et al. 2011).

Calculating the faecal egg count reduction test

The nematicides/anticoccidial effects of the various treatments were done using faecal egg count reduction test (FECRT) as contained in the equation below (Dash et al. 1988),

$$\text{FECRT} = 100 * \left(1 - \left[\frac{T_2}{T_1} \right] \left[\frac{C_1}{C_2} \right] \right)$$

where “T” and “C” are the arithmetic means in the treatment and control groups respectively, while 1 and 2 are pre-treatment and post-treatment, respectively.

Table 1. Prevalence of strongyle infection at pre-treatment according to sex, age group, and location of the sheep.

Risk factors	No. examined	No. positive	Prevalence (%)	OR	95% CI [lower–upper]	P value
Sex						0.9
Male	16	11	68.8	0.92	[0.27–3.19]	
Female	44	31	70.5	Ref		
Age						0.86
Young	21	15	71.4	1.11	[0.35–3.56]	
Adult	39	27	69.2	Ref		
Location						0.788
Nangbagu	20	15	75	1.29	[0.32–5.17]	
Shishegu	20	13	65	0.8	[0.21–3.0]	
Kasalugu	20	14	70	Ref		
Total	60	42	70			

Note: No. = number, Ref = reference, OR = odds ratio, and CI = confidence interval.

Table 2. Prevalence of coccidia infection at pre-treatment according to sex, age group, and location of the sheep.

Risk factors	No. examined	No. positive	Prevalence (%)	OR	95% CI [lower–upper]	P value
Sex						0.812
Male	16	10	62.5	1.15	[0.36–3.74]	
Female	44	26	59.1	Ref		
Age						0.185
Young	21	15	71.4	2.12	[0.69–6.68]	
Adult	39	21	53.8	Ref		
Location						0.788
Nangbagu	20	16	80	4	[0.98–16.27]	
Shishegu	20	10	50	1	[0.29–3.45]	
Kasalugu	20	10	50	Ref		
Total	60	36	60			

Note: No. = number, Ref = reference, OR = odds ratio, and CI = confidence interval.

Statistical analysis

Data were entered into Excel and imported into R version 4.2 for analysis. Descriptive statistics was computed for the means and standard errors of the faecal egg/oocyst counts with regards to the various groups. A univariate logistic regression was done to obtain the effects of the various predictive variables (sex, age, and location) on the prevalence of strongyle and coccidian infection in sheep. We performed a two-way ANOVA to assess the effect of location and treatments on the egg/oocyst counts across the study period. The efficacies of the various treatments against strongyles and coccidia were done using the formula provided above. Multiple comparison of means was done using a Student–Newman–Keuls (SNK) test. Statistical threshold of $P < 0.05$ was set. The results from the analysis are presented in tables and figures.

Ethical approval

The research was reviewed and approved by the University for Development Studies' Institutional Review Board, and in line with the Canadian Council on Animal Care Guidelines, specifically, the care and use of farm animals in research, teaching, and testing.

Results

Approximately, 68.8% and 70.5% of the male and female sheep, respectively examined in this study had at least one species of strongyle. Age, sex, and location of the sheep did not have a significant effect ($P > 0.05$) on the prevalence of strongyle infection (Table 1). The odds of young sheep getting strongyle infection was 11% higher (odds ratio (OR) = 1.11, 95% confidence interval (CI) [0.35–3.56]) than the odds of adult sheep getting strongyle infection. Sex-wise, the odds of male sheep having strongyle infection was 8% lower (OR = 0.92, 95% CI [0.27–3.19]) than the odds of female sheep getting strongyle infection (Table 1).

About 71.4% and 53.8% of the young and adult sheep, respectively examined in this study had at least one species of coccidia. The age, sex, and location of the sheep in the study area did not have a significant effect ($P > 0.05$) on the prevalence of coccidia infection (Table 2). The odds of male sheep having coccidia infection was 15% higher (OR = 1.15, 95% CI [0.36–3.74]) than the odds of female sheep getting coccidia infection (Table 2).

The strongyle FEC of the sheep treated with ABZ reduced from 773.33 ± 201.06 EPG at pre-treatment to 166.67 ± 49.44 EPG at Wk2. Sheep treated with BL10 had pre-treatment strongyle egg count of 713.33 ± 240.01 EPG reduced to

Fig. 1. Average strongyle EPG across the study period.

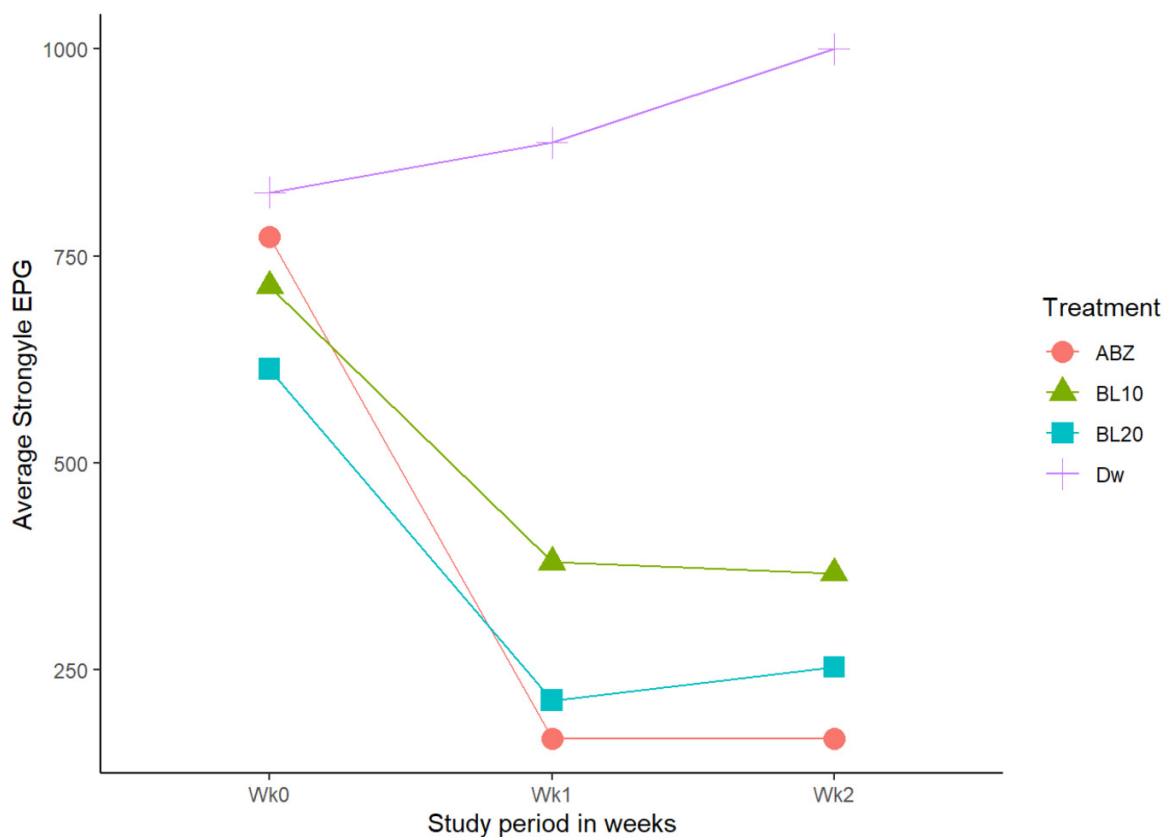


Fig. 2. Average coccidian OPG across the study period.

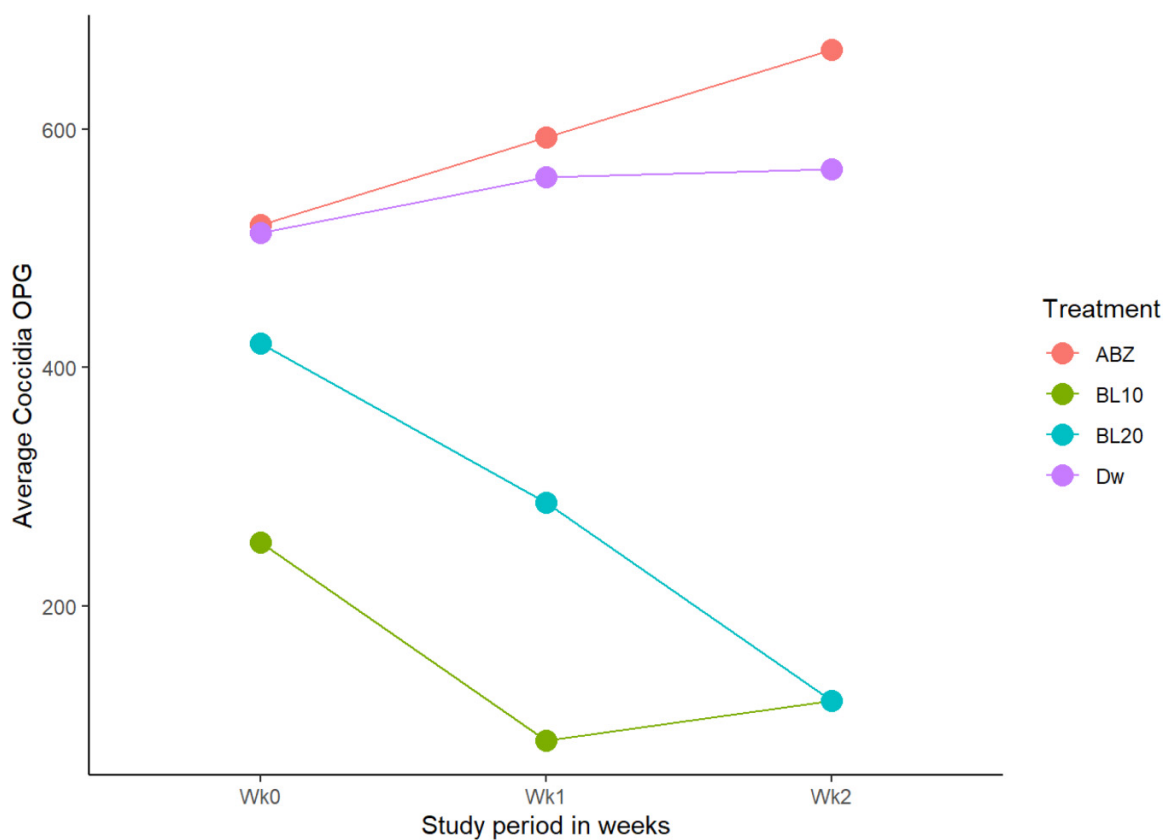


Table 3. Average strongyle EPG base on location and treatment across the study period.

Location	Treatment	Faecal egg count		
		WK0	WK1	WK2
Kasalugu	ABZ	260	20	100
	BL10	550	125	150
	BL20	350	225	225
	DW	1300	1371.4	1557.1
Nangbagu	ABZ	1216.7	283.3	233.3
	BL10	440	200	240
	BL20	966.7	333.3	133.3
	DW	383.3	400	433.3
Shishegu	ABZ	750	175	150
	BL10	1050	700	616.7
	BL20	612.5	162.5	312.5
	DW	500	650	750
P value	Location	0.947	0.737	0.531
	Treatment	0.962	0.182	0.101
	Location × Treatment	0.501	0.676	0.7

366.67 ± 132.26 EPG at Wk2. While sheep treated with BL20 had a strongyle FEC reduced from 613.33 ± 193.68 EPG at pre-treatment to 253.33 ± 100.41 EPG, animals given DW had their strongyle FEC increased from 826.67 ± 447.05 EPG at pre-treatment to 1000 ± 465.99 EPG at Wk2 (Fig. 1).

Animals given ABZ had their coccidia oocyst increased from 520 ± 278.29 OPG at Wk0 to 666.67 ± 284.63 OPG at Wk2. Sheep given BL10 reduced the oocyst count from 253.33 ± 167.58 OPG at pre-treatment to 120 ± 52.73 OPG at 2 weeks post-treatment. Also, BL20 reduced the oocyst count from 420 ± 149.98 OPG at pre-treatment to 120 ± 52.73 OPG at Wk2 post-treatment. Sheep given Dw had their coccidia oocyst increased from 513.33 ± 203.74 OPG at Wk0 to 566.67 ± 228.17 OPG at Wk2 (Fig. 2).

The average strongyle egg count of the sheep given ABZ at Nangbagu was 1216.7 EPG at Wk0 and 233.3 EPG at Wk2. Location and treatment did not have any statistical difference ($P > 0.05$) on the strongyle egg count at Wk0 (Table 3).

The sheep given BL20 at Shishegu had oocyst count of 312.5 OPG at Wk0 and 187.5 OPG at Wk2 (Table 4).

At 2 weeks post-treatment, ABZ had an efficacy of 78.95% against strongyles, while BL10 and BL20 had efficacies of 52.58% and 77.55%, respectively. The efficacies of ABZ, BL10, and BL20 against strongyle were not different ($P = 0.067$). Also, BL10 had an efficacy of 65.08% against coccidia spp., while that of BL20 was 73.03% (Table 5).

Discussion

This study was carried out to determine the efficacy of aqueous extract of *V. amygdalina* leaf against strongyles and coccidia in sheep.

The research brought to light that strongyle prevalence was high in the study area, which is in line with the findings of Ferraz et al. (2015). The higher prevalence of strongyles in

Table 4. Average coccidian OPG base on location and treatment across the study period.

Location	Treatment	Faecal oocyst count		
		Wk0	Wk1	Wk2
Kasalugu	ABZ	20	200	280
	BL10	625	225	50
	BL20	350	525	25
	DW	585.7	657.1	700
Nangbagu	ABZ	1250	1233.3	1283.3
	BL10	60	0	40
	BL20	800	466.7	66.7
	DW	516.7	516.7	500
Shishegu	ABZ	50	125	225
	BL10	166.7	66.7	233.3
	BL20	312.5	100	187.5
	DW	250	350	300
P value	Location	0.212	0.195	0.571
	Treatment	0.763	0.15	0.076
	Location × Treatment	0.247	0.311	0.319

Table 5. Efficacy of the various treatments against strongyles and coccidia spp.

Groups	Treatments	StrongylesFECR (95% CI)	CoccidiaFOCR (95% CI)
A	Albendazole	78.95 (69.21–88.69)	0 ^b
B	BL10	52.58 (43.4–61.76)	65.08 ^a (40.06–90.1)
C	BL20	77.55 (48.88–106.22)	73.03 ^a (36.28–109.78)
D	Dw	—	—
P value		0.067	0.01

Note: The values in the table are the mean faecal egg count reduction test (FECR)/FOCR (95% CI). Values with the same superscripts in the same column are statistically not different from each other. BL20 is 20% bitter leaf extract, BL10 is 10% bitter leaf extract, and Dw is distilled water.

sheep in the study area may be associated with non-frequent deworming of the animals in the study area. The higher prevalence rate of strongyles in female compared to male in this study agrees with the findings of Dagnachew et al. (2011) and Kemal and Minda (2017). This might be due to the physiological status of the animals, as females are more prone to strongyle infections during pregnancy and periparturient period due to stress and decreased immune status. The finding, however, disagrees with that of Abebe et al. (2018), who indicated that strongyles affect both sexes equally.

ABZ had an efficacy of 78.95% with a lower 95% CI of 69.2% against strongyles, using the standard set by the World Association for the Advancement of Veterinary Parasitology, which defines AR as a FECR of less than 95% and (or) a lower 95% CI less than 90%; if only one of these two factors is present, the farm is defined as being “suspected” of having resistance (Wood et al. 1995). Since the efficacy of the ABZ against strongyles in this study meets the conditions, we can say the strongyles are suspected of developing resistance to the ABZ. The findings on the efficacy of ABZ

against strongyle in this study was lower than the 100% efficacy each reported by Menkir et al. (2006) and Kumsa et al. (2013) against strongyles.

The efficacies of *V. amygdalina* leaf extracts in this study against coccidia infection, contradicts that of 35% reported by Al-fifi (2007) in chickens. The difference in the findings may be associated with difference in species with regards to their metabolism. The efficacies of 65.08% and 73.03% for BL10 and BL20, respectively obtained in this study are below that of Adediran et al. (2014), who reported 99.4% efficacy of *V. amygdalina* against coccidian cyst in goats. The difference in this finding could be associated with the difference in the application of the *V. amygdalina* leaf, as Adediran and associates fed the goats with whole bitter leaves as opposed to the aqueous bitter leaf extract given to the sheep in this study.

BL10 and BL20 reduced the strongyle EPG by 52.6% and 77.6%, respectively in this study. This happens to be lower than the 100% efficacy obtained by Adediran et al. (2014) in goats given bitter leaves. The reason for this finding could be due to the way the bitter leaves were used in both studies. This study brought to light that *V. amygdalina* has both anthelmintic and anticoccidial effects in sheep.

The mechanism of anthelmintic activities of the *V. amygdalina* could be comparable to that of benzimidazole, which lead to worms' death by obstructing the microtubule system (Schoenian 2008). Nalule et al. (2013) explained that the feeding ability of the worms given *V. amygdalina* leaves or their extracts suffer paralysis, leading to death as a result of lack of energy.

Vernonia amygdalina leaf extracts showed concentration-dependent as anthelmintic and anticoccidial in sheep; this discovery may be associated with a higher amount of the bioactive compounds found in the higher concentration of the bitter leaf extract (BL20) compared with the relatively lower concentration (BL10).

Conclusions

The study was carried-out to determine the efficacy of aqueous extract of *V. amygdalina* leaf against strongyle and coccidia spp. in sheep. Base on the results obtained, 70% of the sheep in the study area had at least one species of strongyles. BL20 had efficacies of 77.6% and 73.0% against strongyles and coccidia spp., respectively. The efficacy of aqueous extract of *V. amygdalina* leaf against strongyles and coccidia spp. in sheep was concentration-dependent. The use of *V. amygdalina* leaf against strongyles and coccidia spp. in sheep will lead to improved productivity, and the growth of the sheep industry to meet the ever increasing demand for sheep products.

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Data availability

The data set used or analyzed during the current study is available from the corresponding author upon reasonable request.

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Writing – review and editing: IB, ARII, MA, VAN, JK

Competing interests

The authors declare no competing interest for this work.

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